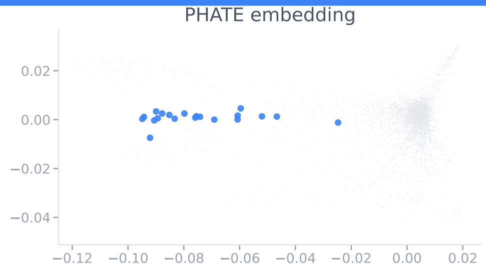
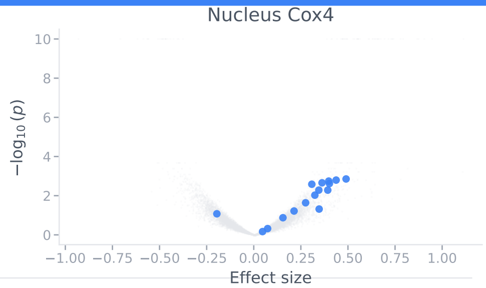
Ribosome biogenesis and SSU processome assembly

Cluster 13





Summary

This cluster shows HIGH confidence for ribosome biogenesis, specifically SSU processome assembly and pre-18S rRNA processing. 15/20 genes (75%) are established SSU processome components or ribosomal proteins with well-documented roles. The morphological phenotypes likely reflect nucleolar disruption and altered ribosome production affecting cell size, shape, and organelle distribution. LOC10272415...

Established genes

UTP18, FCF1, DCAF13, FBL, WDR3, WDR36, NOP56, UTP4, UTP6, WDR46, MPHOSPH10, KRR1, UTP3, RPS16, RPL27

Novel & uncharacterized genes

NEPRO *novel*

Characterized primarily in Notch signaling and neural progenitor maintenance. Clustering with ribosome biogenesis genes is unexpected but may reflect indirect effects on cell proliferation or

NUFIP1 *novel*

RNA-binding protein with limited functional characterization beyond RNA binding capacity. Clustering suggests involvement in ribosome biogenesis, potentially as an RNA chaperone in SSU processome

DUSP11 *novel*

Known RNA 5'-triphosphatase/diphosphatase involved in nuclear mRNA metabolism. Clustering with ribosome biogenesis machinery suggests potential novel role in pre-rRNA processing or modification.

SUPT4H1 *novel*

Well-characterized as component of DSIF complex regulating RNA Pol II transcription elongation and mRNA processing. However, clustering with SSU processome genes suggests potential novel role in

LOC102724159 *unc.*

No functional annotation provided. Clustering with ribosome biogenesis genes suggests potential role in SSU processome or ribosomal RNA processing, but completely unstudied. High discovery potential

15 established · 4 novel · 1 uncharacterized